YASH RAJ

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EDUCATION

ABV-Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
B.Tech in Mathematics & Scientific Computing Expected Graduation: 2029
CGPA: 8.93 / 10.0
Reliance Foundation Scholar
Relevant Coursework: Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, SQL
Frameworks & Libraries: FastAPI, Pandas, NumPy, SciPy, Matplotlib, Seaborn
Tools & Databases: Git, SQLite
Core Competencies: REST API Development, Statistical Optimization, Data Analysis, Backend Engineering

PROJECTS

Automated Wealth & Portfolio Optimization API | *FastAPI, SciPy, Pandas, SQLite*

- Built a FastAPI microservice that applies SciPy SLSQP optimization to maximize portfolio Sharpe Ratio, achieving an optimized Sharpe Ratio of 1.4158 across a 10-asset portfolio using live market data from the FMP API.
- Implemented dynamic risk-free rate lookup from real-time treasury data and persisted portfolio configurations and results using SQLite for reproducible backtesting.

Dynamic Pricing Optimization Engine | *FastAPI, SciPy, PyArrow, Pandas*

- Developed a FastAPI-based pricing engine that runs SciPy optimization over a dataset of 100,000 records to generate profit-maximizing price recommendations.
- Used PyArrow for efficient columnar data processing and built Pandas-based transformation pipelines to prepare large-scale pricing data for optimization.

E-Commerce Customer RFM Segmentation API | *Python, Pandas, FastAPI, SQL*

- Designed and deployed a Recency-Frequency-Monetary (RFM) segmentation pipeline analyzing 100,000+ customer transactions from a real-world Brazilian e-commerce dataset.
- Exposed segmentation outputs through REST API endpoints, enabling integration with downstream analytics and marketing systems.

ACHIEVEMENTS & EXTRACURRICULAR

- Reliance Foundation Scholar: Selected as 1 of 25 scholars nationally for academic merit.
- WorldQuant International Quant Championship (IQC) 2026: Represented university as part of a 4-member team; advanced past Stage 1 by developing and submitting alpha strategies on the WorldQuant BRAIN platform.
- Maintains a public GitHub portfolio of deployable, production-style data science APIs.